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Chapter 3

Subnetting and VLSM

Both the Cisco ICND1 and the ICND2 certification exams expect you to have a sound knowledge of subnetting and variable length subnet masks. Make sure you understand subnetting before attempting this chapter so that you can get the most out of these questions and apply your knowledge. When you're familiar with IP addressing and subnetting, then dive into this chapter and have lots of fun with it as each question presents you with a mathematical challenge!

The Problems You'll Work On

In this chapter, you'll review questions concerning the following topics:

- Determining the number of subnet bits to use to create the desired number of subnets
- Determining the number of host bits required to support the desired number of systems on a subnet
- Calculating the network ID, broadcast address, and range of valid addresses based on given criteria
- Determining the optimal subnet mask to use with the goal of reserving address space

What to Watch Out For

Don't let common mistakes trip you up; watch for the following when working with these questions:

- To calculate the number of networks, the formula is $2^{\text{subnet bits}}$ = number of networks. For example, if you have three subnet bits, the formula gives you $2^3 = 8$ subnets. To calculate the number of systems supported on a network, use this formula: $2^{\text{host bits}} - 2$ = number of valid addresses.
- When solving subnetting questions, always determine the increment value and then calculate all the network IDs. Once you have all the network IDs calculated, determine the network to which the question applies, and then determine the answer. For example, if asked for the third valid address or the broadcast address, determine the network ID first and then figure out the third valid address or the broadcast address.
- Variable length subnet masks (VLSM) do not follow the concept of classes, but instead determine how many host bits are needed to support the number of hosts on the subnet. VLSM allows for each subnet to have a different subnet mask as each subnet supports a different number of hosts.

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Subnetting

86–119 Choose the best answer(s).

86. You are designing the IPv4 address scheme for the network. How do you calculate the broadcast address of a subnet?

(A) Set all host bits to 0

(B) Set all host bits to 1

(C) Set all network bits to 0

(D) Set all subnet bits to 1

87. Your manager would like you to subnet the 129.65.0.0 network into six different networks. What is your new subnet mask?

(A) 255.224.0.0

(B) 255.255.192.0

(C) 255.255.224.0

(D) 255.255.255.224

88. What is the subnet mask of 135.44.33.22/20?

(A) 255.255.192.0

(B) 255.255.240.0

(C) 255.255.224.0

(D) 255.255.248.0

89. You are designing the IP address scheme for your network and you need to subnet the 142.65.0.0 network into 12 different networks. What is your new subnet mask?

(A) 255.255.240.0

(B) 255.255.192.0

(C) 255.255.224.0

(D) 255.255.248.0

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90. How many host bits are needed in order to support 1,010 systems on the network?

(A) 12

(B) 10

(C) 11

(D) 14

91. How many masked bits are needed in order to create at least 20 subnets?

(A) 3

(B) 4

(C) 5

(D) 6

92. You are designing your IPv4 address scheme. How do you calculate the network ID of a subnet?

(A) Set all host bits to 0.

(B) Set all host bits to 1.

(C) Set all network bits to 1.

(D) Set all subnet bits to 0.

93. How many hosts can exist on a network with the address of 180.45.10.20/20?

(A) 1,024

(B) 4,096

(C) 4,094

(D) 1,022

94. How many subnet bits do you need to create 6 networks?

(A) 3

(B) 4

(C) 5

(D) 6

95. Which option on the Cisco router indicates that you can use the first and last subnets of a subnetted network?

(A) VLSM

(B) ip subnet-zero

(C) RIPv1

(D) RIPv2

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96. Which of the following subnet masks matched to the CIDR notation is correct?

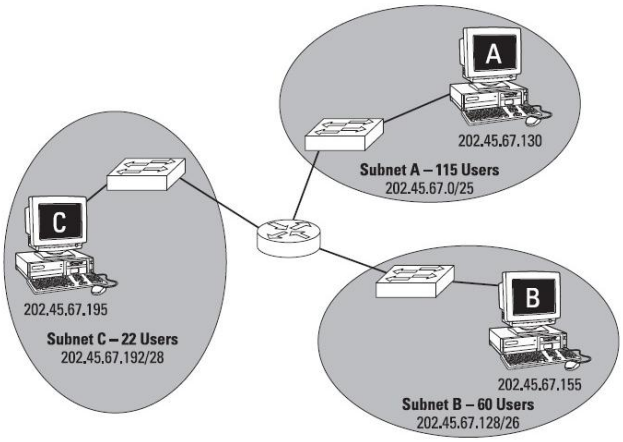
(A) 255.255.255.224 = /27

(B) 255.255.192.0 = /19

(C) 255.255.240.0 = /27

(D) 255.255.255.192 = /28

97. You are designing a network with subnets of variable sizes. In the figure below, identify two problems with the design.



(A) Subnet A does not have enough addresses.

(B) Subnet B does not have enough addresses.

(C) Subnet C does not have enough addresses.

(D) Computer A is assigned an incorrect address.

(E) Computer B is assigned an incorrect address.

(F) Computer C is assigned an incorrect address.

98. Your system uses the IP address of 145.68.23.45/25. How many systems are supported on your network?

(A) 126

(B) 128

(C) 64

(D) 62

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99. The Fast Ethernet port on a router is assigned the IP address of 131.107.16.1/20. How many hosts can exist on the subnet?

(A) 4,096

(B) 4,094

(C) 1,048,574

(D) 1,022

100. You want to assign your serial interface the third valid IP address in the second subnet of 192.168.2.0/26. Which IP address would you use?

(A) 192.168.2.3

(B) 192.168.2.35

(C) 192.168.2.64

(D) 192.168.2.67

101. You are designing the IP address scheme for a branch office that needs to support 92 systems on the network. What should your subnet mask be?

(A) 255.255.255.240

(B) 255.255.255.128

(C) 255.255.255.248

(D) 255.255.255.192

102. What is the minimum number of host bits needed to support 510 hosts on the network?

(A) 12

(B) 7

(C) 9

(D) 14

103. You wish to subnet 137.15.0.0 in order to support 8,190 hosts on each network. You wish to use the smallest subnet size to support this number of hosts. What will be the new subnet mask?

(A) 255.255.255.224

(B) 255.255.192.0

(C) 255.255.240.0

(D) 255.255.224.0

104. How many hosts can exist on a network with the address of 201.10.20.30/27?

(A) 28

(B) 30

(C) 32

(D) 64

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105. With the goal of preserving addresses, which of the following subnet masks would you use to create a subnet of up to 60 hosts?

(A) 255.255.255.224

(B) 255.255.255.192

(C) 255.255.255.252

(D) 255.255.255.248

106. You have a system whose IP address is correctly configured for 198.45.6.87/27. You are now responsible for assigning an IP address to the router. Which of the following IP addresses should be assigned to the router?

(A) 198.45.6.95

(B) 198.45.6.62

(C) 198.45.6.64

(D) 198.45.6.94

107. What is the network ID for the third subnet of 220.55.66.0/27?

(A) 220.55.66.32

(B) 220.55.66.96

(C) 220.55.66.60

(D) 220.55.66.64

108. The 24.60.32.20/11 address is located in which of the following subnets?

(A) 24.32.0.0

(B) 24.64.0.0

(C) 24.96.0.0

(D) 24.16.0.0

109. You need to assign the fourth valid IP address of 107.53.100.10/12 to the Fast Ethernet port on the router. What address would you use?

(A) 107.53.0.4

(B) 107.53.100.4

(C) 107.0.0.4

(D) 107.48.0.4

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110. A network administrator is working with the 195.34.56.0 network, which has been subnetted into 8 networks. Which of the following two addresses can be assigned to hosts on the same subnet? (Select two.)

(A) 195.34.56.35

(B) 195.34.56.14

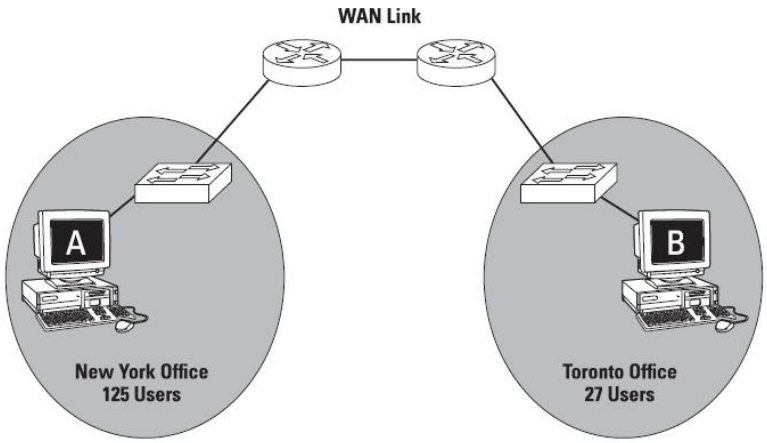
(C) 195.34.56.76

(D) 195.34.56.58

(E) 195.34.56.98

(F) 195.34.56.129

111. Using the network diagram shown in the figure below and having the goal of making the best use of your IP address space, which of the following represents the address scheme you would apply to the Toronto office?



(A) 199.11.33.160/31

(B) 199.11.33.0/25

(C) 199.11.33.128/27

(D) 199.11.33.0/31

(E) 199.11.33.160/30

(F) 199.11.33.128/28

112. What is the network ID of 190.53.159.150/22?

(A) 190.53.132.0

(B) 190.53.144.0

(C) 190.53.152.0

(D) 190.53.156.0

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113. Which of the following statements are true of 205.56.34.53/28? (Choose three.)

(A) The network is not subnetted.

(B) The first valid address of the subnet is 205.56.34.26.

(C) The network ID of the subnet is 205.56.34.48.

(D) The first valid address of the subnet is 205.56.34.65.

(E) The last valid address of the subnet is 205.56.34.55.

(F) The network is subnetted.

(G) The last valid address of the subnet is 205.56.34.47.

(H) The broadcast address of the subnet is 205.56.34.63.

114. What is the broadcast address of 107.50.100.10/12?

(A) 107.63.255.255

(B) 107.50.255.255

(C) 107.48.255.255

(D) 107.53.156.0

115. You need to assign the last valid IP address of 199.45.67.139/26 to the Serial port on your router. What address would you use?

(A) 199.45.67.128

(B) 199.45.67.126

(C) 199.45.67.191

(D) 199.45.67.190

116. What is the last valid address for the subnet of 220.55.66.64/27?

(A) 220.55.66.32

(B) 220.55.66.94

(C) 220.55.66.62

(D) 220.55.66.64

117. Which of the following addresses exists on the same network, knowing that all systems have a subnet mask of 255.255.255.240? (Select two.)

(A) 195.34.56.7.14

(B) 195.34.56.7.30

(C) 195.34.56.7.38

(D) 195.34.56.7.55

(E) 195.34.56.7.17

(F) 195.34.56.7.69

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118. Which of the following statements are true of 190.88.122.45/19? (Choose three.)

(A) The network is not subnetted.

(B) The first valid address of the subnet is 190.88.122.1.

(C) The network ID of the subnet is 190.88.122.0.

(D) The first valid address of the subnet is 190.88.96.1.

(E) The last valid address of the subnet is 190.88.96.254.

(F) The network is subnetted.

(G) The last valid address of the subnet is 190.88.255.254.

(H) The broadcast address of the subnet is 190.88.127.255.

119. Which of the following subnet masks would you typically use on a WAN interface to make the best use of address space?

(A) 255.255.255.224

(B) 255.255.255.192

(C) 255.255.255.252

(D) 255.255.255.248

VLSM

120–123 Choose the best answer(s).

120. Which IP feature allows you to use different subnet masks on different subnets within the network?

(A) IP Subnet Zero

(B) RIPv1

(C) VLSM

(D) RIPv2

121. You are connecting two of your office locations together over a WAN link. Each office has a router with an IP address assigned. What should your subnet mask be in order to leverage the best use of address space?

(A) 255.255.255.0

(B) 255.255.255.192

(C) 255.255.255.224

(D) 255.255.255.252

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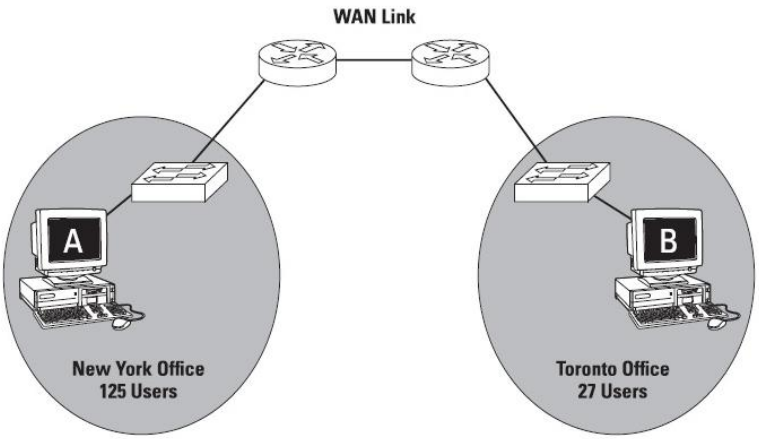
122. Your organization network diagram is shown in the figure below. Your company has the class C address range of 199.11.33.0. You need to subnet the address into three subnets and make the best use of the available address space. Which of the following represents the address scheme you would apply to the New York office?



(A) 199.11.33.160/31

(B) 199.11.33.0/25

(C) 199.11.33.128/27

(D) 199.11.33.0/31

(E) 199.11.33.160/30

(F) 199.11.33.128/28

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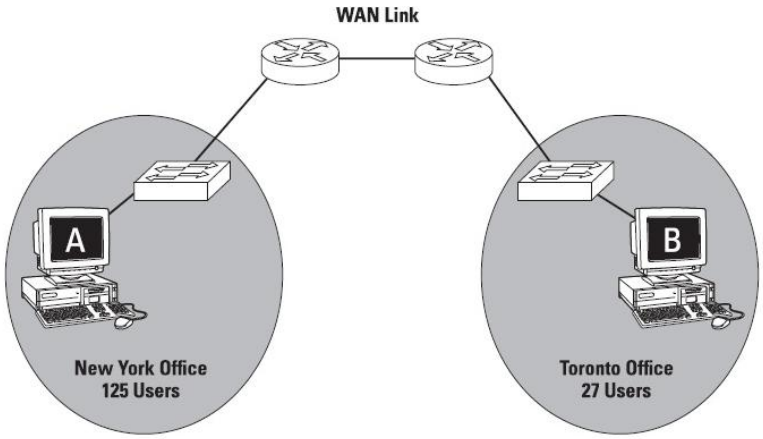
123. Using the network diagram shown in the figure below and having the goal of making the best use of your IP address space, which of the following represents the address scheme you would apply to the WAN link?



(A) 199.11.33.160/31

(B) 199.11.33.0/25

(C) 199.11.33.128/27

(D) 199.11.33.0/31

(E) 199.11.33.160/30

(F) 199.11.33.128/28